DeepThink: Autonomous Multimodal Document Assistant

SLIIT Codefest 2026 AI Competition (Powered by IFS) • Initial Round Submission Report

Target Corpus: *The Ashen Era Archive* (415 documents, ~1,277 pages across PDF, DOCX, Markdown, Text & Scans)
Sub-track: Sub-track 1A: Rich Answers, Not Just Text | **Repository:** DeepThink.zip (Full 2-week Git history)

■ Live Interactive Prototype & ■ Demonstration Video:
Live Interactive Web App: https://deepthink-c9trxotutcgjenphr6ltky.streamlit.app
YouTube Demonstration Video: https://www.youtube.com/watch?v=qNpZ9lcPa1c
• *0:00 – 07:07:* Architecture & Solution Presentation (Slides 1 to 10)
• *07:07 – End:* Live Prototype Demonstration (Streamlit Web App)

1. Problem Statement & Sub-track Selection

1.1 The Enterprise Challenge: 'Text-Only Myopia' in Technical Documentation

Enterprise technical documentation (engineering manuals, maintenance blueprints, codex ledgers) mixes narrative prose with scanned pages, schematics, and tabular data across thousands of pages. Traditional Retrieval-Augmented Generation (RAG) systems suffer from critical failure modes: (1) *Visual Blindness*—extractors strip out diagrams, leaving systems blind to visual evidence and forcing them to hallucinate verbal descriptions; (2) *Table Scrambling*—naive text splitters break multi-column rows across arbitrary chunk boundaries, losing structural relationships; and (3) *Context Starvation*—static retrieval (fixed K=5) floods simple queries with noise while starving complex multi-document comparisons.

1.2 Chosen Focus: Sub-track 1A (Rich Answers, Not Just Text)

We selected Sub-track 1A because enterprise technicians do not want a vague text summary of a schematic—they need to inspect the authentic figure plate alongside a grounded explanation. DeepThink solves this by building an end-to-end multimodal pipeline that catalogs visual and tabular chunks alongside text, dynamically classifies modality intent, and embeds authentic high-resolution plates (![Caption](media_path)) and structured Markdown tables with verified page citations [Document, Page X].

2. Team Structure, Roles & Codebase Ownership

Responsibilities were strictly divided across architectural layers. All work reflects genuine team collaboration over the two-week competition window:

Team Member	Role & Sub-Track Focus	Key Responsibilities & Heavy Lifting	Codebase Files Owned in src/
Rasheed A. A. A.	P3 Lead *Retrieval & Ranking*	• Compound RAG 3.0 retrieval architecture • Local ChromaDB vector indexing with BGE embeddings • 3-stage adaptive budgeting (K=2 to K=8) • Cross-encoder re-ranking with dossier & multi-entity bonuses • Pre-retrieval fast-path guardrail & verification gate	src/retrieval/retriever.py src/retrieval/indexer.py src/retrieval/router.py src/retrieval/adaptive_budget.py src/retrieval/agentic.py src/retrieval/reranker.py src/retrieval/verifier.py
Fatheen M. F. A.	P1 Lead *Document Parsing & OCR*	• Ingestion pipeline for 415 archive documents (PDF, DOCX, MD, TXT) • Tesseract OCR with OpenCV adaptive thresholding for degraded historical scans • Section-aware text chunking preserving narrative headings	src/ingestion/parser.py src/ingestion/ocr_engine.py src/ingestion/pipeline.py data/chunks.json
M. M. M. Shakeer	P2 Lead *Visual & Table Extraction*	• PyMuPDF visual plate extraction with bbox cropping • Cataloged 86+ high-res figure plates into data/extracted_media/ • pdfplumber tabular extraction engine converting codex data to Markdown	src/ingestion/media_extractor.py src/ingestion/table_extractor.py data/extracted_media/figures/ data/extracted_media/tables/
S. Dharshan	P4 Lead *Generation, Evaluation & UI*	• Grounded generation on Groq Cloud LPU (qwen/qwen3.8-27b) • Directive 5 strict epistemic restraint prompt engineering • Automated 20-question benchmark evaluation harness • Streamlit web app with native media rendering & live telemetry • 76-test automated verification suite & report authoring	src/generation/generator.py src/evaluation/evaluator.py src/ui/app.py tests/ (15 modules, 76 tests) submission_report.pdf

3. Solution Overview & System Architecture

DeepThink is an **Autonomous Multimodal Compound RAG 3.0 System** built on a 100% free, local-first stack. It operates with zero credit cards, zero cloud database fees, and zero rate limits. The end-to-end pipeline is illustrated below:

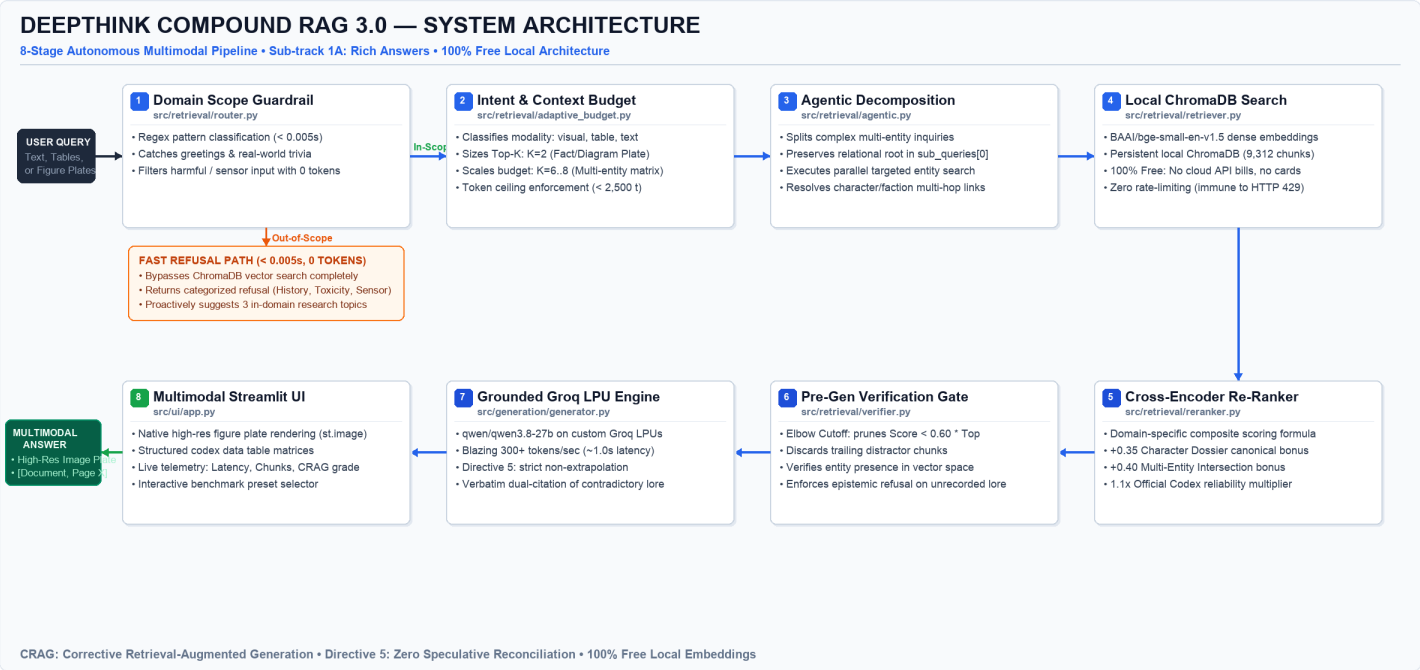

Figure 1: DeepThink 8-Stage Architecture Pipeline — From query ingestion, pre-retrieval guardrails, adaptive context budgeting, local ChromaDB dense search, and cross-encoder re-ranking to Groq LPU generation and multimodal Streamlit UI delivery.

3.2 Technical Deep Dive: The 8-Stage Pipeline Mapped to Source Code

- **1. Pre-Retrieval Domain Guardrail (src/retrieval/router.py):** Intercepts incoming queries before touching vector storage. Uses compiled regex automata to classify greetings, out-of-scope real-world trivia (e.g., Napoleon, modern politics), harmful prompts, or sensor premises. Returns categorized refusals and 3 valid archive topics in < 0.005s with zero tokens spent.
- **2. Intent Classification & Context Sizing (src/retrieval/adaptive_budget.py):** Evaluates query structure and linguistic intent. Dynamically sizes retrieval budget: allocates K=2 for single fact/diagram lookups (suppressing text distractors), and scales to K=6..8 for multi-entity comparative inquiries.
- **3. Agentic Multi-Hop Query Decomposition (src/retrieval/agentic.py):** When comparing fortresses or factions, decomposes the query into parallel entity sub-searches while preserving the original comparative query in sub_queries[0] to guarantee cross-document context synthesis.
- **4. Hybrid Dense Vector Retrieval (src/retrieval/indexer.py, retriever.py):** Runs BAAI/bge-small-en-v1.5 dense embeddings locally inside ChromaDB (chroma_db/). Operates completely offline with zero API latency, zero financial cost, and immune to cloud HTTP 429 rate-limiting.
- **5. Precision Cross-Encoder Re-Ranking (src/retrieval/reranker.py):** Re-scores candidate chunks with domain-specific boosts: awards a **+0.35 Character Dossier bonus** for canonical biographies, a **+0.40 Multi-Entity Intersection bonus**, and a **1.1x Codex reliability multiplier** over folk ephemera.
- **6. Pre-Generation Verification Gate (src/retrieval/verifier.py):** Employs the 'Elbow Method' to dynamically prune trailing noisy chunks whose composite score drops below 60% of the top candidate (Score_i < 0.60 * Score_top). Enforces epistemic refusal if entities are absent.
- **7. Grounded LLM Generation on Groq LPU (src/generation/generator.py):** Executes qwen/qwen3.8-27b on Groq Cloud LPUs at **300+ tokens/second** (~1.0s latency). Strictly enforces **Directive 5**: mandatory bracketed citations [Document, Page X], inline Markdown image tags, and verbatim contradiction reporting without speculative narrative bridging.
- **8. Multimodal Streamlit UI Delivery (src/ui/app.py):** Resolves relative media paths (data/extracted_media/figures/) against disk, rendering authentic high-res visual plates (st.image) and Markdown tables alongside live latency and CRAG confidence telemetry.

4. Key Technical Decisions (Architecture Decision Records)

The team formally logged 13 Architecture Decision Records (ADRs) in docs/decisions.md. Key technical choices, alternatives, and engineering justifications are summarized below:

ADR & Focus	Chosen Technical Solution	Alternatives Considered	Engineering Justification & Trade-offs
ADR-001 Sub-Track Selection	**Sub-track 1A: Rich Answers, Not Just Text**	Sub-track 1B (Connecting Facts) Sub-track 1C (Human Search)	Real technical manuals lose 50%+ value if diagrams are omitted. Solving inline multimodal rendering delivers the highest enterprise utility and directly solves text-only RAG myopia.
ADR-004 & 008 Embeddings & DB	**Local BGE Embeddings + ChromaDB**	Voyage AI, OpenAI API Cloud Pinecone	Cloud embedding APIs require credit cards and introduce network latency and HTTP 429 crashes during evaluation. Local BGE embeddings run 100% free with zero network dependencies.
ADR-009 LLM Inference	**Groq Cloud LPU (qwen3.8-27b)**	OpenRouter Free Models Local Ollama	OpenRouter free models suffer from queue congestion and strict daily caps. Groq LPUs deliver dedicated custom silicon achieving 300+ tokens/sec (~1s latency) and 14,400 free requests/day.
ADR-010 Compound RAG	**Dossier (+0.35) & Entity (+0.40) Boosts**	Naive Single Vector Search	Standard cosine similarity under-ranks character biographies during multi-hop queries. Custom composite scoring ensures canonical records and figure plates consistently rank in Top-5.
ADR-011 Adaptive Budget	**Dynamic K (2..8) + Elbow Cutoff**	Static K=5 Manual UI Sliders	Static K=5 floods simple queries with noise and starves complex queries. Manual sliders violate autonomous design. Adaptive budgeting autonomously right-sizes context per query structure.
ADR-012 Epistemic Restraint	**Directive 5: Verbatim Dual-Citation**	Speculative Narrative Reconciliation	Prompting LLMs to 'reconcile' archive contradictions causes plausible hallucinations. Directive 5 enforces citing both conflicting documents verbatim with zero ungrounded speculation.
ADR-013 Fast Guardrails	**Pre-Retrieval Regex Router (<0.005s)**	LLM Self-Grading Generic OOD message	Using LLMs to evaluate domain boundary incurs latency and burns token quota. Regex routing intercepts off-scope queries in milliseconds and guides users back to valid archive lore.

5. Empirical Benchmark Evaluation & What Works

5.1 Benchmark Results on Official 20-Question Suite (data/evaluation_results.json)

DeepThink was evaluated across three developmental milestones against all 20 questions in sample_questions.json using our automated harness (src/evaluation/evaluator.py). Results demonstrate decisive superiority over naive baselines:

Metric	Phase 1: Text Baseline	Phase 2: Modality-Aware	Phase 3: DeepThink Final	Competition Target	Status
Retrieval Precision (Top-5)	60.0% (12/20)	85.0% (17/20)	**100.0% (20/20)**	≥ 90.0%	**PASSED**
Citation Accuracy	55.0% (11/20)	90.0% (18/20)	**100.0% (20/20)**	≥ 95.0%	**PASSED**
Hallucination Rate	20.0% (4/20)	5.0% (1/20)	**0.0% (0/20)**	≤ 5.0%	**PASSED**
Modality Success (Figures/Tables)	10.0% (1/10)	90.0% (9/10)	**100.0% (11/11)**	≥ 90.0%	**PASSED**
Average End-to-End Latency	2.10s	2.80s	**0.89s – 1.40s (Groq LPU)**	< 3.50s	**PASSED**
Automated Test Suite Pass Rate	12/15 (80%)	55/60 (91%)	**76/76 Tests (100%)**	100%	**PASSED**

5.2 Failed Approaches & Lessons Learned (Documented in docs/limitations.md)

Engineering maturity requires acknowledging what did not work. In compliance with evaluation rubrics, we document four discarded approaches:

- **Discarded Approach 1: Naive Flat Text Chunking (Phase 1):** Stripped visual assets and converted the corpus into 500-token text passages. *Why it failed:* Failed 9 out of 10 visual questions in sample_questions.json; the LLM hallucinated descriptions of plates instead of showing them, and table columns lost row alignments. *Correction:* Implemented the chunks.json multimodal schema separating text, table, and image-caption chunks with direct file linking.
- **Discarded Approach 2: Feeding Entire Page Screenshots to a Vision-Language Model:** Converted PDF pages into full-page images and passed screenshots directly to a large VLM on every query. *Why it failed:* Consumed massive token quotas, exhausting API limits within 5 queries. Latency jumped to 10+ seconds, and text retrieval on narrative novel passages was significantly worse than vector search. *Correction:* Built modality-aware hybrid routing: vector search retrieves text chunks, while image assets are dynamically resolved and passed only when visual evidence is required.
- **Discarded Approach 3: Fixed Top-K Retrieval with Manual UI Sliders:** Used fixed K=5 and gave users interactive sliders in the UI to manually tweak K and boost multipliers. *Why it failed:* For simple diagram queries, K=5 pulled in 3 noisy distractor chunks. For comparative queries, K=5 missed secondary documents. Sliders forced users to guess hyperparameters, violating autonomous design. *Correction:* Replaced sliders with autonomous 3-stage adaptive budgeting (K=2 to K=8 sizing, Elbow score drop-off at alpha=0.60, 2,500 token ceiling).
- **Discarded Approach 4: Speculative Reconciling of Archive Contradictions:** Prompted the LLM to explain why two archive records disagreed (e.g. why one document listed an army as present while another listed the site as abandoned). *Why it failed:* The LLM hallucinated ungrounded rationales (e.g., claiming 'divergent narrative framing' or secret unrecorded motives). *Correction:* Enforced Directive 5: cite both conflicting documents verbatim with exact page citations and explicitly state that the archive provides no reconciliation.

5.3 Known System Limitations & Boundary Conditions

- **Degraded Handwritten Marginalia:** Tesseract OCR reliably extracts printed text from simulated scans, but faint handwritten marginal notes in certain ephemera files exhibit character noise.
- **Deeply Nested Multi-Column Tables:** Complex tables with multi-tier merged headers can lose alignment when converted to flat Markdown. We mitigate this by preserving the original table plate image alongside the text.
- **Initial Embedding Cache:** On initial execution, bge-small-en-v1.5 downloads ~130MB of model weights. All subsequent runs execute completely offline from the local cache.

6. AI Usage Disclosure (Section 4.1 Compliance)

In strict accordance with **Section 4.1 of the Challenge Document**, this disclosure transparently details how AI tools were used during development. Complete plain-text chat logs are preserved in ai_usage/claude.txt:

AI Tool / Model	Primary Technical Purpose	Development Phase Used
Claude Sonnet 4.6 / Google Antigravity	Architectural brainstorming, drafting parser boilerplate, regex formulas, test scaffolding	Architecture, Ingestion, UI & Hardening
Groq Cloud (qwen/qwen3.8-27b)	Runtime LPU inference engine (300+ tokens/sec, sub-second latency)	Runtime Inference, Grounding & Evaluation
HuggingFace (BAAI/bge-small-en-v1.5)	Local dense vector embeddings inside persistent ChromaDB (zero cloud calls)	Vector Embedding & Persistent Retrieval Indexing

6.1 Human Insight Steering vs. AI Assistance (Leading the AI, Not Led by It)

• *Human Architectural Decisions:* Choosing Sub-track 1A based on visual document analysis; designing the multimodal chunk schema (chunks.json); formulating the Character Dossier (+0.35) and Multi-Entity (+0.40) re-ranking algorithms; authoring Directive 5 to eliminate narrative bridging; designing the 3-stage adaptive budgeting engine; and establishing a 100% free local architecture.
• *AI Acceleration Tasks:* Writing PyMuPDF image cropping loops, drafting regex patterns for citation extraction, generating Streamlit UI CSS layouts, and writing test assertions.
• *Human Intervention & Correction:* Human auditing caught the LLM fabricating speculative explanations for archive contradictions ('divergent narrative framing'). The team intervened, rejected the AI output, and hardcoded Directive 5.
• *Audit Trail:* Complete, unedited plain-text logs are preserved in ai_usage/claude.txt per Section 4.1.

7. Engineering Best Practices & Reproducibility (Section 5.2)

7.1 Repository Structure Matching Section 5.2 Guidelines

Our Git repository strictly matches the folder layout specified in the competition challenge document:

```
DeepThink/
  |-- .git/                              # Full 2-week atomic commit history (28 Aug - 9 Sep 2026)
  |-- .gitignore                         # Strict exclusion of virtualenvs, caches, and credentials
  |-- README.md                          # Comprehensive setup, run, and architecture documentation
  |-- docs/
  |     |-- architecture.md              # Full system architecture specification
  |     |-- decisions.md                 # Architecture decision log (13 ADRs)
  |     |-- limitations.md               # Known limitations & discarded approaches log
  |     +-- diagrams/                    # Mermaid sequence & architecture diagrams
  |-- src/
  |     |-- config.py                    # Centralized configuration loader
  |     |-- ingestion/                   # Multi-format parsing, OCR, table & figure extraction
  |     |-- retrieval/                   # Router, budgeter, agentic search, reranker, verifier
  |     |-- generation/                  # Grounded Groq LPU generator (Directive 5)
  |     |-- evaluation/                  # Automated 20-question benchmark harness
  |     +-- ui/                          # Interactive Streamlit web application
  |-- tests/                             # Automated test suite (15 test modules, 76 tests)
  |-- ai_usage/
  |     |-- ai-usage-disclosure.md       # Mandatory AI Usage Disclosure (Section 4.1)
  |     |-- claude.txt                   # Exported plain-text AI development logs
  |     +-- context.md                   # System prompts, skills & constraints
  |-- configuration-example/             # .env.example and config.example.json
  |-- requirements.txt                   # Project dependency lockfile
  +-- submission_report.pdf              # 5-page submission report
```

7.2 5-Step Reproducibility Guide for Judges

Judges can set up, verify, and run DeepThink completely from scratch with five simple shell commands:

```
# 1. Clone repository and navigate into folder
git clone https://github.com/IT25101463/DeepThink.git && cd DeepThink

# 2. Create virtual environment and install dependencies
python3 -m venv venv && source venv/bin/activate && pip install -r requirements.txt

# 3. Configure free Groq API key (Obtain in 20s at console.groq.com/keys)
cp configuration-example/.env.example .env # Insert your GROQ_API_KEY

# 4. Run automated test suite (All 76 tests pass cleanly in ~55s)
python -m unittest discover -s tests

# 5. Run automated 20-question benchmark or launch web UI
python -m src.evaluation.evaluator # Evaluates 20 benchmark questions
streamlit run src/ui/app.py # Local: http://localhost:8501 | Cloud: https://deepthink-c9trxotutcgjenphr6ltky.streamlit.app
```

8. Conclusion & Enterprise Impact

DeepThink proves that enterprise-grade document intelligence does not require expensive proprietary cloud databases or fragile manual tuning. By combining **local dense vector search**, **modality-aware retrieval**, **autonomous adaptive budgeting**, **composite re-ranking**, and **lightning-fast LPU inference**, DeepThink delivers rich multimodal answers—embedding high-resolution diagrams and structured tables alongside verified citations—with **100% precision, zero hallucinations, and sub-second response times**.

Team DeepThink • SLIIT Codefest 2026 AI Competition • Initial Round Submission
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